Software Requirements Specification (SRS)

Cover

Field	Value
Project	Department Course Timetable Management System
Document	Software Requirements Specification
Version	v1
Date	10.11.2026
Team number	Team 03
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Repository	https://github.com/tuna645/seng271-team03-timetable

Revision History

Version	Date	Author(s)	Summary of changes
v1	10.11.2026	Tuna Eser, Levent Gürel	Initial requirements from the project description.

1. Introduction

1.1 Purpose

This document describes the software requirements for the Department Course Timetable Management System. It explains what the system will do and defines the rules and constraints for scheduling course sessions.

1.2 Scope

- **In scope:**
  - Reading input data from CSV and TXT files (courses, instructors, rooms, constraints).
  - Automatically creating a weekly timetable for 4 year groups (Years 1 to 4).
  - Checking hard constraints (preventing room, instructor, and year overlaps).
  - Checking soft constraints and showing warnings for preferences.
  - Showing the timetable on the console as a grid.
  - Letting the user manually move or delete an assignment.
  - Creating a classroom request email draft for sessions with no room and saving it to a text file.
- **Out of scope:**
  - Databases (SQL).
  - Graphical user interface (GUI) or web interface.
  - Sending real emails automatically over the internet.
  - Individual student-level course enrollments and elective conflicts.

1.3 Definitions

Term	Definition
Session	A theory block (-T) or lab block (-L) of a course that must be scheduled in consecutive hours.
Time slot	A one-hour period between 09:00 and 17:00 on weekdays (e.g., 09:00–10:00).
Year group	The study year of students (Year 1, 2, 3, or 4).
Hard constraint	A mandatory rule that cannot be broken. If a rule is broken, the assignment is rejected.
Soft constraint	A preference that should be satisfied if possible. If broken, the system only shows a warning.
Special request	A specific rule for a course, such as a fixed time or a required room.
Unscheduled session	A session that could not be placed because of a room problem or a constraint conflict.

1.4 References

- SENG 271 Software Project Assignment: Department Course Timetable Management System.
- SENG 271 Term Project Guide.
- SENG 271 Tutorial.

2. Overall Description

2.1 Product Perspective

The system is a console-based Java application. It reads input files from the data/ folder, processes them to generate a weekly timetable, displays the results on the console, and saves reports and email drafts to the output/ folder.

2.2 User

User	Main goals	Characteristics
Department Course Timetable Coordinator	Create weekly schedules for the department, check conflicts, change hours manually, and request classrooms when rooms are full.	Department staff member who organizes courses and classrooms.

2.3 Operating Environment

- **Language:** Java (JDK 21 or 25).
- **Environment:** Terminal / Command Line / IntelliJ IDEA on Windows and macOS.

2.4 Assumptions and Constraints

- Working days are Monday to Friday, 09:00 to 17:00 (8 one-hour slots per day).
- Each course belongs to one year group. Mandatory courses of the same year cannot overlap.
- Theory and lab hours are separate blocks and must be scheduled in consecutive hours on the same day.
- Multi-hour sessions cannot be split between different days or non-consecutive slots.

3. Functional Requirements

ID	Group	Requirement	Priority	Source	Screen

FR-01	Data loading	The system shall read course data from a CSV file (data/courses.csv).	Must	§3.7	Screen 2
FR-02	Data loading	The system shall read instructor information and availability from a text file (data/instructors.txt).	Must	§3.7	Screen 2
FR-03	Data loading	The system shall read classroom and lab data from a CSV file (data/rooms.csv).	Must	§3.7	Screen 2
FR-04	Data loading	The system shall read special requests from a text file (data/constraints.txt).	Must	§3.7	Screen 2
FR-05	Data loading	If an input line contains invalid data (such as letters in a numeric field), the system shall skip that line, show a warning with the line number, and continue reading the rest of the file.	Must	§3.7	Screen 2
FR-06	Data loading	If an input file references an unknown course, instructor, or room, the system shall ignore that record and print a warning message.	Must	§3.7	Screen 2

FR-07	Automatic generation	The system shall place fixed-time courses first before scheduling other sessions.	Must	§3.8	Screen 3
FR-08	Automatic generation	The system shall place remaining sessions into valid days, slots, and rooms that satisfy all hard constraints.	Must	§3.8	Screen 3
FR-09	Conflict handling	The system shall not assign a session to a slot or room if it violates any hard constraint (HC-01 to HC-09).	Must	§3.5.1	Screen 3
FR-10	Conflict handling	If a soft constraint cannot be satisfied, the system shall still place the session and show a warning message.	Should	§3.5.2	Screen 3
FR-11	Conflict handling	If a session cannot be placed, the system shall mark it as unscheduled and record whether it is a room shortage (NO ROOM) or a constraint conflict (CONSTRAINT).	Must	§3.5, §3.8	Screen 5
FR-12	Timetable viewing	The system shall display the weekly timetable for a selected year group as a table grid on the console.	Must	§3.9, §3.11	Screen 4

FR-13	Timetable viewing	The system shall display a list of all scheduled sessions with their assigned time and room.	Must	§3.9, §3.11	Screen 1
FR-14	Timetable viewing	The system shall display a list of all unscheduled sessions and soft-constraint warnings.	Must	§3.9, §3.11	Screen 5
FR-15	Manual modification	The system shall allow the user to change the day, start time, and room of a scheduled session.	Must	§3.9	Screen 6
FR-16	Manual modification	If a manual change violates a hard constraint, the system shall reject the change and keep the current assignment.	Must	§3.9	Screen 6
FR-17	Manual modification	The system shall allow the user to remove a session assignment from the timetable.	Must	§3.9	Screen 1
FR-18	Classroom request	When a session has no suitable room, the system shall generate a formal request email draft showing the course, student count, required room type, and suggested time.	Must	§3.10	Screen 7
FR-19	Classroom request	The system shall save the generated email draft to a file	Must	§3.10	Screen 7

		in the output/ folder when the user confirms.			
FR-20	Output	The system shall allow the user to save the full timetable to a text or CSV file.	Must	§3.11	Screen 1

4. Scheduling Constraints

4.1 Hard Constraints (Must Never Be Violated)

ID	Constraint	Rule in our own words	Data needed to check it	Message on violation
HC-01	Instructor conflict	An instructor cannot teach two classes at the same time.	Instructor ID, Day, TimeSlot	[FAIL] Instructor conflict: Instructor is already teaching another course.
HC-02	Room conflict	A room cannot be used by more than one session at the same time.	Room ID, Day, TimeSlot	[FAIL] Room conflict: Room is already occupied at this time.
HC-03	Year-group conflict	Two courses of the same year group cannot be at the same time.	Year group, Day, TimeSlot	[FAIL] Year conflict: Year group already has a session at this time.

HC-04	Working hours	Sessions must be between 09:00 and 17:00, Monday to Friday.	Day, Start Time, Hours	[FAIL] Working hours: Session must end by 17:00.
HC-05	Room capacity	A room's capacity must be greater than or equal to the student count.	Room capacity, Course student count	[FAIL] HC-05: Room capacity is smaller than student count.
HC-06	Room type	Theory sessions must be in Classrooms, and lab sessions must be in Laboratories.	Session type (-T/-L), Room type	[FAIL] HC-06: Room type does not match session type.
HC-07	Instructor availability	A session cannot be scheduled when the instructor is unavailable.	Instructor ID, Unavailable slots	[FAIL] HC-07: Instructor is unavailable at this time.
HC-08	Fixed-time requirements	Courses with a fixed time request must be placed at that exact time.	Fixed time request, Current schedule	[FAIL] HC-08: Fixed-time request cannot be satisfied.
HC-09	Room availability	A session cannot be scheduled when the room is unavailable.	Room ID, Unavailable periods	[FAIL] HC-09: Room is unavailable during this slot.

4.2 Soft Constraints (Preferences)

ID	Constraint	How our system considers it	Warning shown
SC-01	Theory before laboratory	The system tries to schedule the theory block before the lab block.	[WARN] SC-01: Laboratory session is scheduled before theoretical session.
SC-02	Instructor preferences	The system checks if the instructor prefers morning hours and tries morning slots first.	[WARN] SC-02: Instructor prefers morning hours; session placed in afternoon.
SC-03	Avoiding unnecessary gaps	The system tries to avoid gaps of 3 or more empty hours between classes for students.	[WARN] SC-03: Year group has a 3-hour gap on this day.
SC-04	Consecutive teaching hours	Multi-hour sessions are placed in consecutive hours.	[WARN] SC-04: Multi-hour session is not consecutive.

4.3 Special Scheduling Requests

Request type	Example	Mandatory or preferred	How the system handles it
Fixed time	SENG301 must be on	Mandatory	Scheduled in Step 1 before other courses. If conflicting, marked as CONSTRAINT failure.

	Wednesday, 13:00–16:00		
Specific room	SENG401 must use Laboratory B202	Mandatory	The system only checks room B202 for this session.
Instructor unavailable day	SENG201 instructor unavailable on Friday	Mandatory	The system does not test Friday slots for this instructor.
Laboratory before theory	SENG302 lab may precede theory	Preferred	Disables the SC-01 warning for this course.

5. Input Data Requirements

File	Format	Fields (in order)	Example line	Validation rules
data/courses.csv	CSV	code,title,year,instructorId,theoryHours,labHours,students	SENG201,Object-Oriented Programming,2,I02,3,2,55	year is 1–4; hours are 0 or more; students > 0; instructorIdexists.

data/instructors.txt	TXT	id,name,unavailableSlots,preferredSlots	I01,Dr. Kaya,Mon 09:00-12:00,morning	id is not empty; day and hours are valid slot format.
data/rooms.csv	CSV	id,type,capacity	A101,Classroom,80	type is Classroom or Laboratory; capacity > 0.
data/constraints.txt	TXT	courseCode,requestType,details,priority	SENG301,FIXED_TIME,Wed 13:00-16:00,MANDATORY	courseCodeexists; priority is MANDATORY or PREFERRED.

Input Error Handling

Error	Example	System reaction
Missing file	rooms.csv not found	Prints "Error: File not found: data/rooms.csv" and returns to the menu.
Invalid numeric value	"three" for theory hours	Skips the line and prints "courses.csv, line 8: invalid value 'three' -> line skipped".

Incomplete record	A line missing a column	Skips the line and prints "Error on line X: missing field -> record skipped".
Reference to nonexistent item	Request for SENG499	Ignores the request and prints "Warning: Course SENG499 not found".

6. Output Requirements

Output	Content	Shown on screen	Saved to file
Weekly timetable per year group	Grid with days (Mon-Fri) and hours (09:00-17:00), showing course, room, and instructor.	Yes (Screen 4)	Yes (e.g. output/timetable_year2.txt)
All scheduled sessions	List of placed sessions with course name, day, time, and room.	Yes (Option 4)	Yes
Unscheduled sessions	Sessions that could not be scheduled, problem type	Yes (Screen 5)	Yes

	(NO ROOM or CONSTRAINT), and reason.		
Warnings	List of soft-constraint warnings (SC-01 to SC-04).	Yes (Screen 5)	Yes
Classroom request email draft	Letter draft with course code, student number, required room type, and suggested time.	Yes (Screen 7)	Yes (output/classroom_request_SENG201.txt)

7. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Reliability	The system shall not crash when an input file has wrong data or when a file is missing.
NFR-02	Usability	If the user enters an invalid menu option, the system shall show an error message and redisplay the menu.
NFR-03	Performance	The system shall generate the timetable for the sample data within 5 seconds.

NFR-04	Maintainability	The code shall be organized into separate Java classes with clear responsibilities.

8. Changes from the Previous Version

(Empty in v1).

AI-Use Declaration

Tool	Used for	How we checked the result
ChatGPT / Gemini	Analyzing project requirements from the assignment PDF and clarifying document expectations.	Reviewed all extracted requirements against the official project guide as a team to ensure complete alignment.
ChatGPT / Gemini	Understanding the required Git repository layout and project folder structure (docs/, data/, output/, src/).	Cross-checked the directory tree directly against Section 10 of the Project Guide and Tutorial Step 4.1.
ChatGPT / Gemini	Checking scheduling algorithm logic for potential flaws and conflict scenarios.	Hand-simulated sample scheduling cases using pen and paper to verify constraint logic.
ChatGPT / Gemini	Learning Java language syntax, standard library features, and OOP best practices.	Tested code snippets independently in IntelliJ IDEA to ensure we fully understand every statement.
Grammarly / AI Assistant	Proofreading English grammar, spelling, and phrasing in project documentation.	Both team members read through all sections to ensure clarity and appropriate terminology.